Stanley Wang

stanley.wang.cs@gmail.com | U.S. & Canada | GitHub | LinkedIn | Portfolio

Skills

Languages: Python, TypeScript, Java, Go, C++, C, JavaScript, MySQL, OCaml, Lua, Bash
Technologies: Git, Unix/Linux, React, Flask, REST APIs, WebSocket, Redis, PyTorch, AWS, Docker, CI/CD, NGINX

Education

McGill University, Bachelor of Science, Computer Science Sep 2022 – Dec 2026

Experience

Software Engineering Intern, **Electronic Arts**, EADP Commerce, Redwood City, CA May 2026 – Aug 2026
- Prototyped an **agentic commerce** surface that stages and verifies personalized purchases on the Madden Mobile Web Store
- Implemented **cross-boundary identity handoff** across identity and commerce microservices
- Engineered agent runtime with **FastAPI** and **OpenAI Agents SDK**: SSE streaming, dynamic tooling, **SQLite** purchase-plan store
- Projected **$157K** (**+4.65%**) annual revenue lift, enabled in-part by designing and shipping upstream **multi-item cart** support
- Created and deployed an **agent orchestration** system on an internal server, which integrated with team's tooling and platforms

Software Engineering Intern, **Autodesk**, Montreal, QC May 2025 – Aug 2025
- Contributed to **Flow Production Tracking** and **Creative Review's** public beta, enhancing **UI/UX** and **performance** for major studios (**Disney**, **Sony**, **MAPPA**)
- Introduced a **visual debugging tool** with **user event sequence tracing**; improved deployment reliability via **TDD** and **CI/CD**

Software Engineering Intern, **BETA Technologies**, Montreal, QC — Case Study Jan 2025 – Apr 2025
- Developed structural analysis library in **Python**, reducing **3-day** manual process to **10 seconds**
- Engineered a batch-processor to efficiently modify **multi-gigabyte** finite element aircraft models
- Accelerated future development of structural analysis tools by **95%**

Lead Researcher, **McGill AI Ethics Lab**, Montreal, QC May 2024 – Dec 2024
- Designed a **cross-language API** library in **Go** with **C bindings** for **Python** integration, improving team velocity by **40%**

Projects

Eastwatch, Agent Orchestration GitHub
- Built an issue-board-driven **agent orchestrator** turning GitLab/GitHub/Jira/Obsidian board gestures into detached, resumable **tmux** workers running **Claude** and **Pi** sub-agents
- Designed **reconciliation-loop** architecture (**launchd**) with fleet TUI, crash-detected resumable runs, and multi-provider **model routing**; scaled to a hosted multi-user **Docker** variant (**SQLite** outbox dispatch, isolated per-user containers) for an EA team

Mini-C Compiler, Compilers
- Built an **end-to-end compiler** for a **C subset** in **Java** — hand-written **lexer** and **Pratt Parser** down to **MIPS32** assembly
- Implemented **semantic analysis** — **name analysis** over scoped **symbol tables** and **type checking** — over a generated **AST**
- Implemented **graph-coloring register allocation** over a **control-flow graph**, with **backward liveness analysis** and **spill** code
- Extended the language with **classes**, single **inheritance**, and **virtual dispatch**; verified with a **JUnit** fixture suite on **MARS**

OS Kernel Simulator, Systems Programming
- Built an **OS simulator** in **C**: a shell loads script **programs** into simulated memory and runs them as concurrent processes
- Implemented preemptive **round robin** scheduling with **aging**, raising process priority so long jobs never starve
- Simulated **demand paging** with per-process **page tables**, evicting the **LRU** frame on **page faults**

Pi Talk, Open-Source Agentic Tooling GitHub
- Released an **open-source** voice layer for **Claude Code** and **Pi**, a beloved **agentic harness**, via **streaming OpenAI TTS**
- Shipped the port in a day **orchestrating AI agents** over spec-locked tickets: 88 unit tests, zero live **API** calls

Trading Fours, Full Stack & Machine Learning Demo Video | GitHub
- Built a music recommendation engine that predicts genre with **XGBoost**; deployed on **AWS EC2** with **Docker** and **CI/CD**